

Forder — Book IV

Proposition 14

Formal Reconstruction — Technical Documentation

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CGJTEAMLAB

If A, B, C, D lie on a circle with centre O , A, O, C are collinear, and B, D lie on opposite sides of AC , then the opposite angles BAD and BCD are supplementary.

1 Introduction

Forder IV.14 is the first proposition in the present circle sequence whose printed proof is genuinely short while the formal proof architecture is substantial.

Let A, B, C, D lie on one circle with centre K . Assume that AC is a diameter and that B and D lie on opposite sides of the diameter line. Forder's conclusion is

$$\angle BAD + \angle BCD = \pi.$$

In the production formalization this is expressed synthetically: the side BC is extended through C to a point X , and one proves

$$\angle BAD \cong \angle XCD.$$

Thus $\angle BAD$ is congruent to the angle supplementary to $\angle BCD$.

The public theorem is

```
hilbert_IV14_diameter_supplement
```

in

```
CGJteamLab/Proposition4_14.lean
```

The proof depends directly on IV.13, but it also requires substantial reusable circle/order infrastructure from `HilbertInterfaceIV`. In particular, the formal development must prove an order fact that the source diagram leaves implicit: the chord BD meets the diameter AC at a point lying strictly inside the diameter.

2 Classical Configuration

Let AC be a diameter of the circle with centre K . The points B and D lie on opposite sides of the diameter line.

B, D on opposite sides of AC

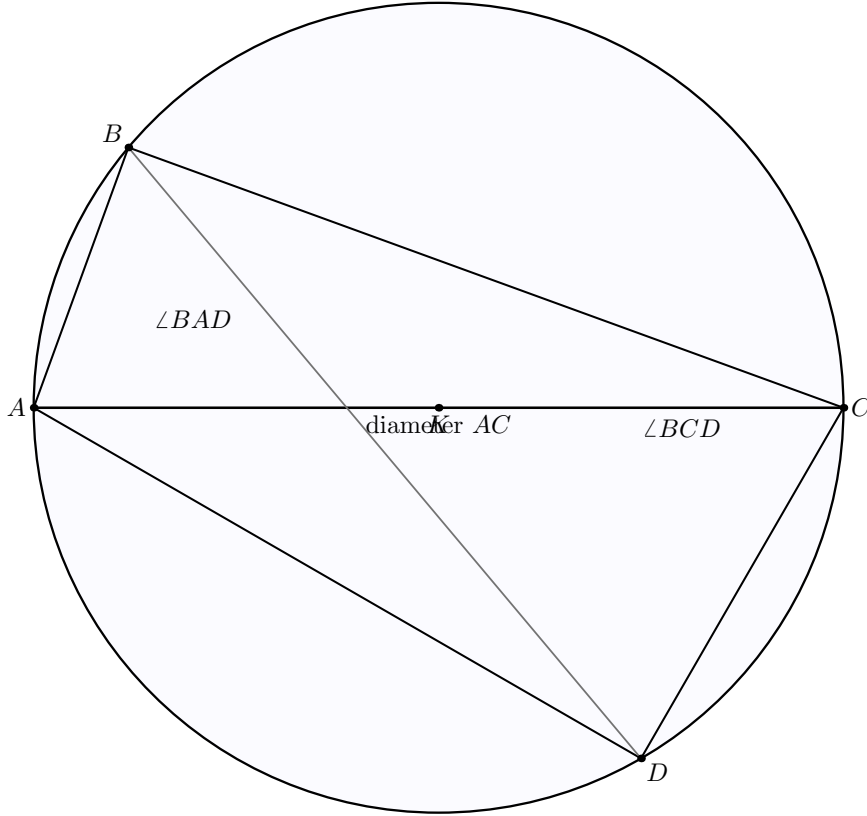


Figure 1: Forder IV.14. The diameter is AC ; the circle points B, D lie on opposite sides of its carrier. The theorem asserts that $\angle BAD$ and $\angle BCD$ are supplementary.

The source proof decomposes both target angles by the diameter:

$$\angle BAD = \angle BAC + \angle CAD,$$

and

$$\angle BCD = \angle BCA + \angle ACD.$$

Because AC is a diameter, IV.13 gives the two right angles

$$\angle ABC, \quad \angle ADC.$$

Consequently the two acute-angle pairs satisfy

$$\angle BAC + \angle BCA = \frac{\pi}{2},$$

and

$$\angle CAD + \angle ACD = \frac{\pi}{2}.$$

Adding the decompositions yields

$$\angle BAD + \angle BCD = \pi.$$

This is almost the whole printed proof. The formal reconstruction must replace all three uses of angular arithmetic:

1. the two “sum to a right angle” statements;
2. the decomposition of $\angle BAD$ through the ray AC ;
3. the decomposition of the supplementary angle to $\angle BCD$.

All three are reconstructed from explicit rays, side-of-line information, right-angle congruence, and synthetic angle addition/subtraction.

3 Forder’s Statement and Proof

Forder’s statement is:

If A, B, C, D lie on a circle with centre O , A, O, C are collinear, and B, D lie on opposite sides of AC , then the angles BAD and BCD are supplementary.

His proof invokes IV.13 twice. The angles subtending the diameter AC at B and D are right. Therefore the two acute angles of triangle ABC fill one right angle and the two acute angles of triangle ADC fill another.

The fact that B, D lie on opposite sides of AC implies that the segments BD and AC meet. The diagram then shows that AC is an interior divider of $\angle BAD$, while CA is an interior divider of $\angle BCD$. Forder therefore combines the two pairs of acute angles and obtains one straight angle [1].

The formalization follows that architecture but cannot use the phrases “the diagram shows” or “add the two equalities”. Those are exactly the places where the proof expands.

4 Formal Statement

IV.14 is slightly different from its neighbours in the production API. There is no separate proposition abbreviation `HilbertForderIV14....`. The public theorem is exported directly:

```
theorem hilbert_IV14_diameter_supplement
  [H : HilbertIncidence Geo]
  [HE : @HilbertEuclideanPlane Geo H]
  (K R A B C D : Geo.Point)
  (diameter : Geo.Line)
  (hAC : Ne A C)
  (hAdiam : H.OnLine A diameter)
  (hCdiam : H.OnLine C diameter)
  (hKdiam : H.OnLine K diameter)
  (hAcircle : HilbertCircle Geo K R A)
  (hBcircle : HilbertCircle Geo K R B)
  (hCcircle : HilbertCircle Geo K R C)
  (hDcircle : HilbertCircle Geo K R D)
  (hOppBD : HilbertOppositeSide Geo B D diameter) :
```

```
exists X : Geo.Point,
  BookZeroSupplement Geo B C D D X /\
  Geo.AngleCongruent B A D X C D
```

The conclusion contains both pieces of the synthetic supplement statement.

First,

```
BookZeroSupplement Geo B C D D X
```

records the straight-line extension $B - C - X$ relative to the angle BCD .

Second,

```
Geo.AngleCongruent B A D X C D
```

states that the angle at A is congruent to that adjacent supplementary angle at C .

Thus the theorem does not assert a numerical equation of angle magnitudes. It constructs the supplement geometry.

5 Production Architecture

The current module imports only

```
CGJteamLab.HilbertInterfaceIV
CGJteamLab.Proposition4_13
```

The proposition-specific proof is still organized by private implementation lemmas. Their stage suffixes are internal historical names, not public API:

```
hilbert_forder_diameter_right_decomposition_stage36
hilbert_forder_IV14_diameter_rays_inside_stage38
hilbert_forder_IV14_prescribed_perpendicular_stage39
hilbert_forder_IV14_perpendicular_inside_supplement_stage40
hilbert_forder_IV14_first_component_stage41
hilbert_forder_IV14_second_component_stage42
hilbert_forder_IV14_diameter_supplement_stage43
```

Two facts discovered during the IV.14 development were promoted to the reusable interface because their statements are proposition-independent:

```
hilbert_right_triangle_third_angle_congruent
hilbert_circle_diameter_chord_intersection_inside
```

The second of these is used directly by the production IV.14 proof. The first is a reusable by-product of the development, but the final `Proposition4_14.lean` proof does not require it

as a local step.

6 Right-Angle Decomposition from IV.13

A recurring local block in IV.14 begins with a diameter AC and a circle point B off the diameter. IV.13 yields

$$\angle ABC \text{ right.}$$

The proof extends AB through B to a point E :

$$A - B - E.$$

The adjacent angle CBE is therefore right as well.

Euclid I.32 is then applied to triangle CAB with AB produced through B . It constructs a point P such that

$$C - P - E$$

and supplies the component congruences

$$\angle ACB \cong \angle CBP, \quad \angle CAB \cong \angle PBE.$$

Hence the right angle CBE is decomposed synthetically into the two acute angles of triangle ABC .

This local theorem replaces the numerical identity

$$\angle ACB + \angle CAB = \frac{\pi}{2}.$$

The important point is methodological: even a familiar “the acute angles of a right triangle are complementary” statement is realized by an explicit exterior construction and two angle congruences.

7 The Hidden Order Lemma: the Chord Crosses the Diameter Internally

From the opposite-side hypothesis, the line segment BD meets the diameter carrier at some point Y :

$$B - Y - D.$$

For the later angle decomposition it is not enough to know that Y lies somewhere on the line AC . One must prove

$$A - Y - C.$$

The reusable theorem is now

```
hilbert_circle_diameter_chord_intersection_inside
```

internal chord-diameter crossing

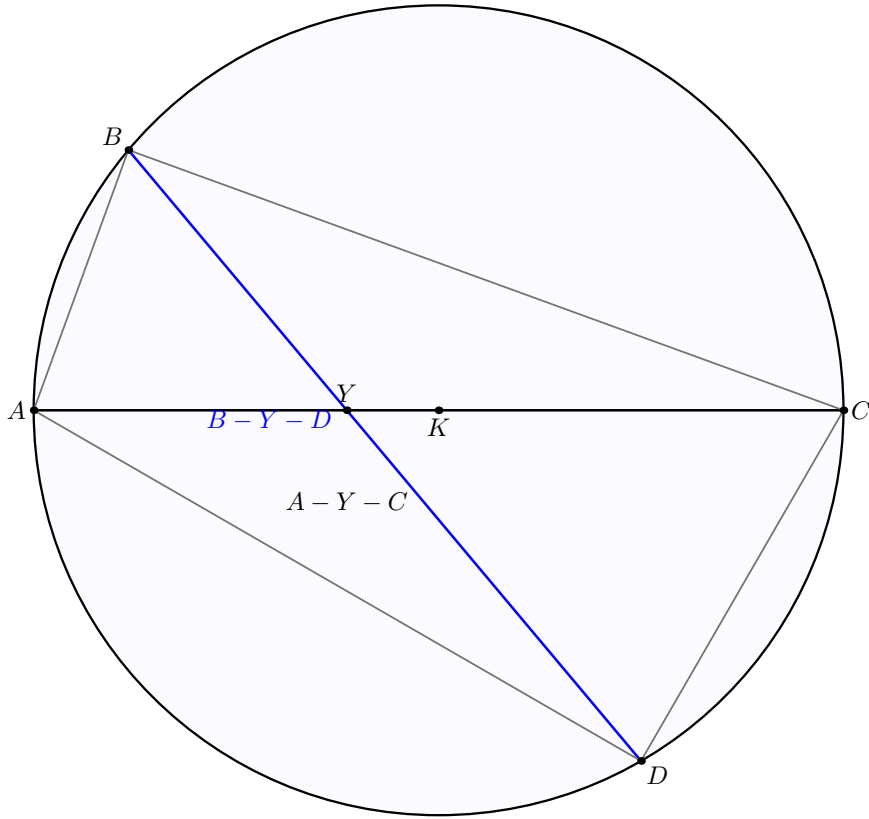


Figure 2: The hidden order fact in IV.14. The opposite-side hypothesis gives a point Y with $B - Y - D$ on the diameter carrier. The formal proof must show the stronger statement $A - Y - C$, so the diameter rays are genuine interior dividers of the two target angles.

Its proof has two branches.

If the centre K lies on chord BD , uniqueness of the intersection of the chord carrier and the diameter carrier forces

$$Y = K.$$

Since K is the midpoint of diameter AC , the conclusion follows.

Otherwise Y is a proper interior point of chord BD . A reusable circle theorem gives

$$KY < KB.$$

All radii are congruent, so

$$KY < KA.$$

If Y lay beyond either endpoint of the diameter, the order geometry would force the reverse strict inequality. Segment-order trichotomy therefore leaves only

$$A - Y - C.$$

This is a particularly good example of formalization exposing a hidden diagrammatic premise. No continuity or coordinate argument is needed; the proof is purely synthetic.

8 From the Crossing Point to Interior Rays

Once

$$B - Y - D, \quad A - Y - C$$

are known, the proposition-specific helper

```
hilbert_forder_IV14_diameter_rays_inside_stage38
```

converts the order configuration into the two statements

$$\overrightarrow{AC} \text{ is interior to } \angle BAD$$

and

$$\overrightarrow{CA} \text{ is interior to } \angle BCD.$$

In Lean these are

```
HilbertRayMeetsSegment Geo A C B D
HilbertRayMeetsSegment Geo C A B D
```

This is exactly the information needed later by the angle-addition API. The theorem contains no angle calculation at all; it is a pure incidence/order bridge between the circle configuration and the angle decomposition.

9 Constructing a Perpendicular on the Prescribed Side

The formal proof next creates a ray CP perpendicular to the diameter at C , specifically on the same side of the diameter as D .

The construction uses IV.13 to obtain the right angle

$$\angle ABC.$$

Hilbert's angle-construction theorem copies this angle to the vertex C with first arm CA , choosing the side that contains D . The resulting point P satisfies

$$P, D \text{ are on the same side of } AC$$

and

$$\angle ACP \text{ is right.}$$

The private production helper is

```
hilbert_forder_IV14_prescribed_perpendicular_stage39
```

This step is important because the eventual supplementary angle to $\angle BCD$ will be decomposed through CP . The point P is not an arbitrary perpendicular witness: its half-plane is prescribed by the location of D .

10 Constructing the Supplementary Ray

Extend BC through C to a point X :

$$B - C - X.$$

This is the point that will appear in the final `BookZeroSupplement` witness.

The proof now needs a precise nesting of three rays at C :

$$CP \text{ inside } \angle ACX,$$

$$CD \text{ inside } \angle ACP,$$

and

$$CP \text{ inside } \angle XCD.$$

These statements are the core output of

```
hilbert_forder_IV14_perpendicular_inside_supplement_stage40
```

The proof is not based on a visual comparison of slopes. It uses strict angle comparison.

For example, Euclid I.16 gives an inequality showing that the right angle copied to ACP is smaller than the larger exterior angle ACX . This places CP inside ACX .

Similarly, in the right triangle ADC , the acute angle at C is strictly smaller than the right angle ACP . This places CD inside ACP .

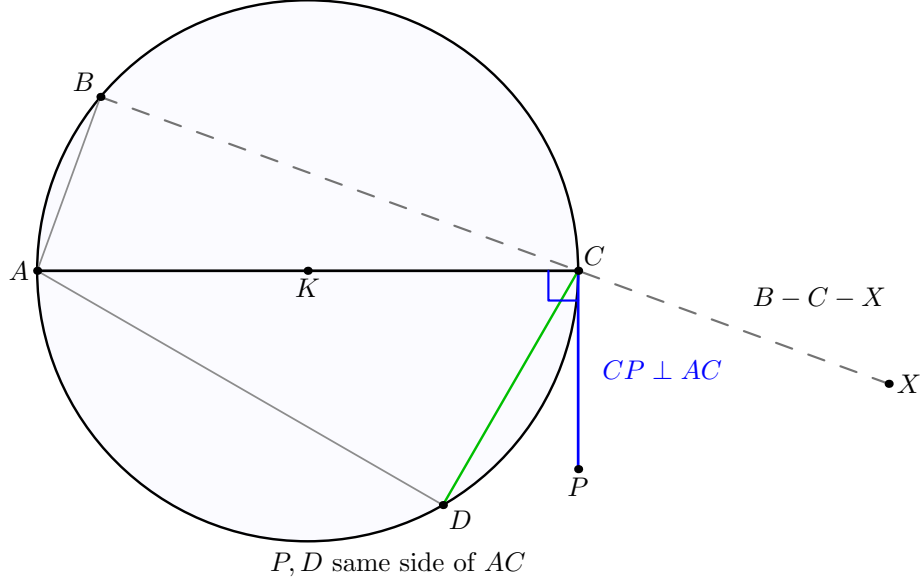


Figure 3: The auxiliary configuration at C . The ray CX extends CB ; the ray CP is perpendicular to the diameter and chosen on the side containing D . The proof establishes the nested order CD inside ACP and CP inside XCD .

Nested-interior-ray lemmas then combine these order relations to obtain the final position of CP inside XCD .

Thus a single source picture expands into a genuine chain of synthetic angle inequalities and ray-order statements.

11 The First Component Congruence

The first target component is

$$\angle BAC \cong \angle XCP.$$

The proof again applies Euclid I.32, now to triangle ABC with BC produced through C to X . It obtains an interior splitter CQ of the exterior angle ACX and congruences

$$\angle BAC \cong \angle ACQ,$$

$$\angle ABC \cong \angle QCX.$$

The angles ABC and ACP are both right, hence congruent. Therefore the right-hand component XCQ of the exterior decomposition is congruent to ACP .

At this point the proof compares two decompositions of congruent whole angles:

$$\angle ACX \quad \text{and} \quad \angle XCA.$$

The corresponding right components are congruent, so synthetic angle subtraction yields

$$\angle ACQ \cong \angle XCP.$$

Combining with I.32 gives

$$\boxed{\angle BAC \cong \angle XCP}.$$

The private helper closing this block is

12 The Second Component Congruence

The second target component is

$$\angle CAD \cong \angle PCD.$$

This time the proof works with triangle ACD .

IV.13 gives

$$\angle ADC \text{ right.}$$

Extend CD through D to E :

$$C - D - E.$$

The adjacent angle

$$\angle ADE$$

is right.

Apply I.32 to triangle ACD with CD produced to E . The construction gives a splitter DQ and

$$\angle CAD \cong \angle ADQ,$$

$$\angle ACD \cong \angle QDE.$$

The whole angle ADE is right, and the auxiliary angle PCA is also right because $CP \perp AC$. Hence the wholes are congruent.

The component EDQ is congruent to ACD . Synthetic subtraction of the corresponding components therefore gives

$$\angle ADQ \cong \angle PCD.$$

Combining with I.32 yields

$$\boxed{\angle CAD \cong \angle PCD}.$$

This is the output of

The two component lemmas mirror one another, but they are not literally the same proof: the first uses the extension $B - C - X$, while the second uses a new extension $C - D - E$.

13 Final Assembly

At A , the ray AC lies inside $\angle BAD$. Hence

$$\angle BAD$$

is decomposed into

$$\angle BAC \quad \text{and} \quad \angle CAD.$$

At C , the ray CP lies inside $\angle XCD$. Hence

$$\angle XCD$$

is decomposed into

$$\angle XCP \text{ and } \angle PCD.$$

The two component congruences proved above are

$$\angle BAC \cong \angle XCP$$

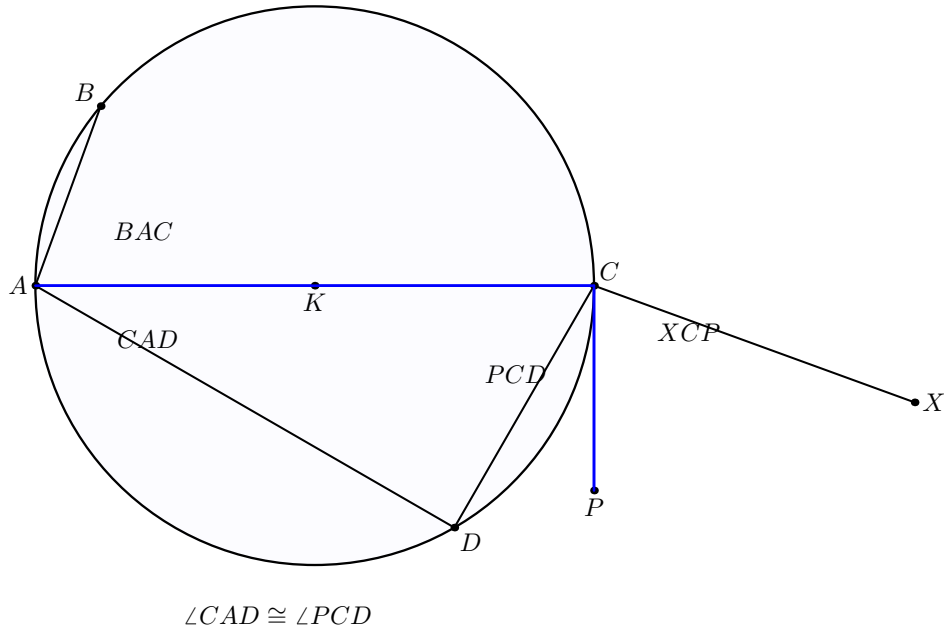
and

$$\angle CAD \cong \angle PCD.$$

Synthetic angle addition therefore gives

$$\angle BAD \cong \angle XCD.$$

$$\angle BAC \cong \angle XCP$$



$$\angle CAD \cong \angle PCD$$

Figure 4: Final component matching in IV.14. The diameter ray AC decomposes $\angle BAD$, while the auxiliary perpendicular CP decomposes the supplementary angle $\angle XCD$. The two corresponding component pairs are proved congruent separately and then reassembled.

Finally, $B - C - X$ is exactly the straight-line extension required by the Book Zero supplement predicate. The theorem returns

$$X$$

together with

$$\text{Supplement}(BCD, DCX)$$

and

$$\angle BAD \cong \angle XCD.$$

This is the synthetic content of Forder's equation

$$\angle BAD + \angle BCD = \pi.$$

14 Lean Formalization

The public theorem is deliberately thin:

```

theorem hilbert_IV14_diameter_supplement
  ...
  : exists X : Geo.Point,
    BookZeroSupplement Geo B C D D X /\
    Geo.AngleCongruent B A D X C D :=
  hilbert_forder_IV14_diameter_supplement_stage43
  ...

```

All proposition-specific detail is private to Proposition4_14.lean.

The proof architecture is:

```

IV.13 right angles
  |
  v
diameter/right-angle decomposition
  |
  +-----+
  |                                     |
  v                                     v
internal chord/diameter      prescribed perpendicular
intersection                  CP
  |                             |
  v                             v
diameter rays interior      extend BC through C to X
  |                             |
  +-----+-----+
  |                                     |
  v                                     v
nested ray configuration
  |
  +-----+-----+
  |                                     |
  v                                     v
first component      second component
BAC ~= XCP           CAD ~= PCD
  |                   |
  +-----+-----+
  |
  v
angle addition
  |
  v
BAD ~= XCD
  |
  v
BookZeroSupplement

```

This is considerably richer than the three-line angle calculation in the source, but every extra step has a specific synthetic role.

15 Reusable Interface Ownership

Two general results associated with this development belong to `HilbertInterfaceIV`.

15.1 Diameter–chord intersection

The theorem

```
hilbert_circle_diameter_chord_intersection_inside
```

states a general fact about a chord connecting opposite sides of a diameter. Its statement does not mention IV.14 and it is reusable independently.

15.2 Third angle in right triangles

The theorem

```
hilbert_right_triangle_third_angle_congruent
```

says that two right triangles with one pair of corresponding acute angles congruent have their remaining acute angles congruent.

It was isolated during the IV.14 development and belongs semantically in the interface. The final production proof of IV.14 uses the more direct decomposition/subtraction route described above, so this theorem is not a hidden proposition-level dependency.

This distinction is important for documentation: provenance during proof development is not the same thing as ownership in the final library.

16 Relation to IV.13

IV.14 imports `Proposition4_13.lean` directly.

IV.13 is used repeatedly:

- to prove the diameter angle at B right;
- to prove the diameter angle at D right;
- to provide a right angle that can be copied to construct the prescribed perpendicular at C .

Thus IV.14 is the first proposition in this block whose production module has an explicit numbered predecessor dependency:

```
HilbertInterfaceIV
Proposition4_13
-> Proposition4_14
```

IV.12 is not required by this proof.

17 Hilbert, Book Zero, and Earlier Euclid Components

17.1 Incidence and plane separation

The opposite-side hypothesis is structural. It supplies the crossing point of chord BD with the diameter line and controls the side on which the copied perpendicular CP is constructed.

17.2 Order

Betweenness appears at every major transition:

$$A - Y - C, \quad B - Y - D, \quad B - C - X, \quad C - D - E.$$

These order relations are converted into interior-ray witnesses used by the angle-decomposition API.

17.3 Congruence

Circle membership supplies congruent radii and IV.13 supplies right angles. The theorem that all right angles are congruent is then used repeatedly to match components across different vertices.

17.4 Hilbert angle construction

The prescribed perpendicular at C is obtained by copying an existing right angle to a specified side of the diameter. This is the synthetic substitute for choosing a coordinate direction perpendicular to AC .

17.5 Euclid I.16 and I.32

I.16 supplies strict angle inequalities needed to prove ray interiority in the nested configuration.

I.32 is used to create explicit decompositions of exterior angles into remote interior components. It is the main bridge from the classical right-triangle picture to the synthetic angle-decomposition machinery.

17.6 Book Zero supplement

The final theorem does not use a numerical straight angle. The supplement is represented by a point X on the continuation of BC , precisely as encoded by `BookZeroSupplement`.

18 Foundational Boundary

IV.14 introduces no new axiom and uses no coordinate geometry.

The difficult-looking step that a chord crossing a diameter does so internally is proved from:

- plane separation,
- segment order,
- circle-radius congruence,
- the theorem that an interior point of a chord lies strictly inside the circle.

The remainder of the proof uses the already established Euclidean-plane angle infrastructure.

The formal statement therefore sits cleanly within the same `HilbertEuclideanPlane` framework as IV.13.

19 Classical and Formal Dependency Structures

Forder’s visible proof is essentially

$$\begin{array}{c}
 AC \text{ diameter} \\
 \downarrow \\
 \angle ABC, \angle ADC \text{ right (IV.13)} \\
 \downarrow \\
 \begin{array}{l}
 \angle BAC + \angle BCA = \frac{\pi}{2} \\
 \angle CAD + \angle ACD = \frac{\pi}{2}
 \end{array} \\
 \downarrow \\
 AC \text{ and } BD \text{ cross} \\
 \downarrow \\
 \angle BAD + \angle BCD = \pi.
 \end{array}$$

The formal reconstruction expands this to

$$\begin{array}{c}
 \text{IV.13} \\
 \downarrow \\
 \text{right-angle decompositions} \\
 \downarrow \\
 \text{hilbert_circle_diameter_chord_intersection_inside} \\
 \downarrow \\
 AC, CA \text{ certified as interior dividing rays} \\
 \downarrow \\
 \text{prescribed perpendicular } CP \\
 \downarrow \\
 B - C - X \text{ and nested interior rays} \\
 \downarrow \\
 \begin{array}{l}
 \angle BAC \cong \angle XCP \\
 \angle CAD \cong \angle PCD
 \end{array} \\
 \downarrow \\
 \text{synthetic angle addition} \\
 \downarrow \\
 \angle BAD \cong \angle XCD \\
 \downarrow \\
 \text{BookZeroSupplement.}
 \end{array}$$

At module level:

```

HilbertInterfaceIV
  |
Proposition4_13
  |
  v
Proposition4_14

```

20 Diagrammatic Audit

Four figures are justified for IV.14.

1. The source configuration shows the theorem itself.
2. The chord–diameter intersection figure isolates the hidden order lemma.
3. The perpendicular/supplement figure explains the auxiliary ray CP and the extension $B - C - X$.
4. The component figure displays the final two-by-two angle matching used by synthetic addition.

These figures correspond to genuinely different logical layers. Combining them into one dense diagram would conceal the formal proof structure.

21 Conclusion

Forder IV.14 is a compact classical argument whose formal reconstruction reveals three distinct kinds of hidden structure.

First, the crossing of BD and AC must be strengthened to an internal crossing of the diameter. Second, the supplementary relation must be realized by an actual extension point X . Third, the angle arithmetic must be replaced by explicit decompositions and componentwise congruence.

The resulting proof remains fully synthetic. It uses IV.13, plane separation, betweenness, angle construction, Euclid I.16 and I.32, and the project’s angle addition/subtraction machinery. The public theorem exposes only the natural geometric conclusion, while the proposition-specific construction remains private to `Proposition4_14.lean`.

This proposition is therefore a good model of the purpose of the present documentation: not merely to record that Lean accepted the theorem, but to show precisely which pieces of geometry are compressed into the classical diagram and how they are recovered in the formal proof.

References

- [1] Henry George Forder, *Foundations of Euclidean Geometry*, Cambridge University Press, Cambridge, 1927.